

Directed numbers: adding and subtracting integers

A Cambridge insert: numbers below zero, the number line as a road, and what a double minus does.

LESSON 2–4

3 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 1.1

LESSON CLOCK

20'

25'

30'

30'

15'

Numbers below zero	20 min
Ordering integers	25 min
Adding	30 min
Subtracting	30 min
Where they are used	15 min

LEARNING OBJECTIVES

- Read and place negative numbers on a number line.
- Compare and order integers.
- Add and subtract integers using the number line and the sign rules.
- Use directed numbers for temperature, height and money.

TEXTBOOK

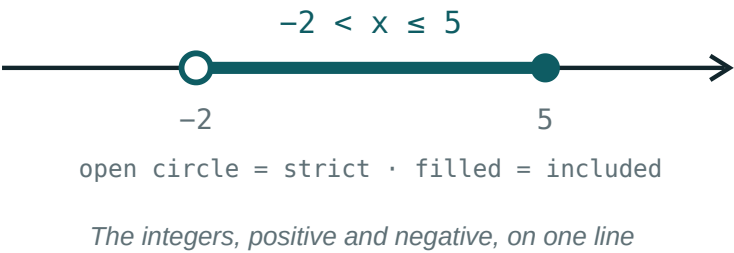
National	pp. 9–14
Cambridge	Stage 7, pp. 8–13
Workbook	Exercise 1.1

01

Explanation

Numbers below zero

Counting numbers stop at zero; the world does not. Temperature, debt, depth and floors below ground all need numbers on the other side.



SITUATION	POSITIVE	NEGATIVE
temperature	above freezing	below freezing
height	above sea level	below sea level
money	saved	owed
a lift	upper floors	the basement

EVERY NUMBER HAS AN OPPOSITE

5 and -5 sit the same distance from zero on opposite sides, and they add to zero. Zero is its own opposite, and is neither positive nor negative.

Ordering integers

On the number line, the number further to the right is the larger. Among negatives that reverses the everyday feeling: -2 is **larger** than -7 .

COMPARISON	TRUE?	WHY
$5 > 3$	yes	further right
$-2 > -7$	yes	-2
$-7 > -2$	no	the digits mislead
$0 > -1$	yes	zero beats every negative
$-100 > 1$	no	every negative is below every positive

THE BIGGER THE DIGITS, THE SMALLER A NEGATIVE NUMBER IS

-100 is far below -2 . Reading the digits and forgetting the sign is the error this whole block is designed to prevent.

Adding

Adding a positive moves right; adding a negative moves left.

SUM	ON THE LINE	ANSWER
$-3 + 5$	from -3 , five right	2
$3 + (-5)$	from 3 , five left	-2
$-3 + (-5)$	from -3 , five left	-8
$-5 + 5$	back to the start	0

SAME SIGNS ADD, DIFFERENT SIGNS SUBTRACT

With the same sign, add the digits and keep the sign. With different signs, subtract the smaller digits from the larger and keep the sign of the larger. The number line proves both.

Subtracting

$$a - b = a + (-b)$$

Subtracting is adding the opposite, so a minus in front of a negative becomes a plus.

EXPRESSION	REWRITTEN	ANSWER
$7 - 10$	$7 + (-10)$	-3
$-4 - 6$	$-4 + (-6)$	-10
$5 - (-3)$	$5 + 3$	8
$-5 - (-3)$	$-5 + 3$	-2

TWO MINUSES NEXT TO EACH OTHER MAKE A PLUS

$5 - (-3)$ is 8 , not 2 . Taking away a debt leaves you better off — which is exactly what the rule says in words.

Where they are used

PROBLEM	WORKING	ANSWER
from -7°C the temperature rises 12°	$-7 + 12$	5°C
from 4°C it falls 9°	$4 - 9$	-5°C
the difference between 6°C and -3°C	$6 - (-3)$	9°
a diver at -12 m rises 5 m	$-12 + 5$	-7 m
a lift from floor -2 up 7	$-2 + 7$	floor 5

A DIFFERENCE IS ALWAYS A SUBTRACTION

From -3° to 6° the rise is $6 - (-3) = 9$ degrees — and the double minus is exactly why the answer is bigger than either number.

02 Worked examples

EXAMPLE 1 Compute $-3 + 5$, $3 + (-5)$ and $-3 + (-5)$.

1 Different signs: $5 - 3 = 2$, sign of the larger — positive.
2

2 Different signs: $5 - 3 = 2$, sign of the larger — negative.
 -2

3 Same signs: add and keep the sign.
 -8

ANSWER

2, -2 , -8

EXAMPLE 2 Compute $5 - (-3)$ and $-5 - (-3)$.

① Subtracting is adding the opposite.

② $5 + 3 = 8$

③ $-5 + 3 = -2$

ANSWER8 and -2 **EXAMPLE 3** The temperature is -7°C and rises by 12° . What is it now?

① A rise means adding.

② $-7 + 12$

③ $= 5^{\circ}\text{C}$

ANSWER 5°C **03 Interactive model**

Draw a vertical number line as a thermometer on the board and keep it there all week; every question in this block is answered by pointing at it.

Signs on the number line

Which is larger, -2 or -7 ?

-2

-7

equal

cannot say

$-3 + 5$ equals:

-8

2

-2

8

$3 + (-5)$ equals:

8

2

-2

-8

$-3 + (-5)$ equals:

-8

-2

2

8

$7 - 10$ equals:

3

-3

17

-17

$5 - (-3)$ equals:

2

8

-8

-2

$-5 - (-3)$ equals:

-8

-2

2

8

From -7° a rise of 12° gives:

5°

-5°

19°

-19°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Integer	Butun son	Целое число
Negative number	Manfiy son	Отрицательное число
Positive number	Musbat son	Положительное число
Number line	Koordinata to'g'ri chizig'i	Числовая прямая
Opposite number	Qarama-qarshi son	Противоположное число
Below zero	Noldan past	Ниже нуля
Rise	Ortish	Повышение
Fall	Kamayish	Понижение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

On the number line the larger number is:

further left

further right

nearer zero

the one with more digits

The opposite of -9 is:

9

-9

0

$\frac{1}{9}$

$-4 - 6$ equals:

2

-2

10

-10

$6 - (-3)$ equals:

3

9

-9

-3

Every negative number is:

above zero

below zero

equal to zero

a fraction

A diver at -12 m who rises 5 m is at:

-17 m

-7 m

7 m

17 m

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-3 + 5$

ANSWER 2

2. $3 + (-5)$

ANSWER -2

3. $-3 + (-5)$

ANSWER -8

4. $7 - 10$

ANSWER -3

5. $-4 - 6$

ANSWER -10

6. $5 - (-3)$

ANSWER 8

7. The opposite of -9

ANSWER 9

Medium · the standard question

7 PROBLEMS

1. $-5 - (-3)$

ANSWER -2

2. Order $-7, 2, -2, 0, 5$

ANSWER $-7 < -2 < 0 < 2 < 5$

3. From -7° a rise of 12°

ANSWER 5°

4. From 4° a fall of 9°

ANSWER -5°

5. The difference between 6° and -3°

ANSWER 9°

6. A diver at -12 m rises 5 m

ANSWER -7 m

7. $-8 + 8$

ANSWER 0

Hard · stretch and reason

7 PROBLEMS

1. $-7 + 12 - (-5)$

ANSWER 10

2. $-3 - (-8) + (-4)$

ANSWER 1

3. The difference between -15° and -4°

ANSWER 11°

4. A lift goes from floor -3 up 9, then down 4

ANSWER floor 2

5. Which is larger, -100 or -1 ?

ANSWER -1

6. Write a subtraction whose answer is -6

ANSWER e.g. $2 - 8$

7. A debt of 50 000 is cancelled: the change in what you own

ANSWER $+50\ 000$

07 Homework

Homework — 5 tasks

Draw a number line beside every question until the signs feel automatic.

1. Compute $-6 + 9$, $6 + (-9)$ and $-6 + (-9)$.
2. Compute $4 - 11$ and $-4 - (-11)$.
3. Order -12 , 3 , -3 , 0 , 8 , -20 from smallest to largest.
4. The temperature falls from 2°C by 11° . What is it now?
5. Find the difference between 8°C and -6°C , and say why the answer is larger than either.